$$
\begin{aligned}
&\mathcal{C}_{qe}{}^{i1\_i2\_i3\_i4\_} \to \\
&\frac{1}{16\pi^2}\left(\frac{1}{216}\,\frac{1}{m_\Phi^2}\left(s_\gamma^2\,\overline{y_e}^{pi3}\,y_e^{pi4}\left(-54\,c_\gamma^2\,\overline{y_u}^{i2r}\,y_u^{i1r}+5\,g_1^2\,\delta_{i1i2}\right)+13\,g_1^2\,c_\gamma^2\,\overline{y_u}^{i2p}\,y_u^{i1p}\,\delta_{i3i4}-\right.\right.\\
&\quad\left.s_\gamma^2\,\overline{y_d}^{i2p}\,y_d^{i1p}\left(54\,c_\gamma^2\,\overline{y_e}^{ri3}\,y_e^{ri4}+5\,g_1^2\,\delta_{i3i4}\right)\right)-\frac{2}{81}\sum_p g_1^4\,\mathrm{LF}_{3,0}\left[m_{\tilde d}^p\right]\delta_{i1i2}\,\delta_{i3i4}+\\
&\frac{5}{108}\sum_p g_1^4\,\mathrm{LF}_{4,-1}\left[m_{\tilde d}^p\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{8}{405}\sum_p g_1^4\,\mathrm{LF}_{5,-2}\left[m_{\tilde d}^p\right]\delta_{i1i2}\,\delta_{i3i4}-\\
&\frac{2}{27}\sum_p g_1^4\,\mathrm{LF}_{3,0}\left[m_{\tilde e}^p\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{5}{36}\sum_p g_1^4\,\mathrm{LF}_{4,-1}\left[m_{\tilde e}^p\right]\delta_{i1i2}\,\delta_{i3i4}-\\
&\frac{8}{135}\sum_p g_1^4\,\mathrm{LF}_{5,-2}\left[m_{\tilde e}^p\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{27}\sum_p g_1^4\,\mathrm{LF}_{3,0}\left[m_{\tilde l}^p\right]\delta_{i1i2}\,\delta_{i3i4}+\\
&\frac{5}{72}\sum_p g_1^4\,\mathrm{LF}_{4,-1}\left[m_{\tilde l}^p\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{4}{135}\sum_p g_1^4\,\mathrm{LF}_{5,-2}\left[m_{\tilde l}^p\right]\delta_{i1i2}\,\delta_{i3i4}-\\
&\frac{1}{81}\sum_p g_1^4\,\mathrm{LF}_{3,0}\left[m_{\tilde q}^p\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{5}{216}\sum_p g_1^4\,\mathrm{LF}_{4,-1}\left[m_{\tilde q}^p\right]\delta_{i1i2}\,\delta_{i3i4}-\\
&\frac{4}{405}\sum_p g_1^4\,\mathrm{LF}_{5,-2}\left[m_{\tilde q}^p\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{8}{81}\sum_p g_1^4\,\mathrm{LF}_{3,0}\left[m_{\tilde u}^p\right]\delta_{i1i2}\,\delta_{i3i4}+\\
&\frac{5}{27}\sum_p g_1^4\,\mathrm{LF}_{4,-1}\left[m_{\tilde u}^p\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{32}{405}\sum_p g_1^4\,\mathrm{LF}_{5,-2}\left[m_{\tilde u}^p\right]\delta_{i1i2}\,\delta_{i3i4}+\\
&\frac{1}{18}\left(s_\gamma^2\,\overline{y_e}^{pi3}\,y_e^{pi4}\left(-9\,c_\gamma^2\,\overline{y_u}^{i2r}\,y_u^{i1r}+g_1^2\,\delta_{i1i2}\right)+4\,g_1^2\,c_\gamma^2\,\overline{y_u}^{i2p}\,y_u^{i1p}\,\delta_{i3i4}-\right.\\
&\quad\left.s_\gamma^2\,\overline{y_d}^{i2p}\,y_d^{i1p}\left(9\,c_\gamma^2\,\overline{y_e}^{ri3}\,y_e^{ri4}+2\,g_1^2\,\delta_{i3i4}\right)\right)\mathrm{LF}_{1,2}[m_\Phi]+\\
&\frac{1}{36}\left(s_\gamma^2\,\overline{y_e}^{pi3}\,y_e^{pi4}\left(9\,c_\gamma^2\,\overline{y_u}^{i2r}\,y_u^{i1r}+g_1^2\,\delta_{i1i2}\right)-3\,g_1^2\,c_\gamma^2\,\overline{y_u}^{i2p}\,y_u^{i1p}\,\delta_{i3i4}+\right.\\
&\quad\left.3\,s_\gamma^2\,\overline{y_d}^{i2p}\,y_d^{i1p}\left(-3\,s_\gamma^2\,\overline{y_e}^{ri3}\,y_e^{ri4}+g_1^2\,\delta_{i3i4}\right)\right)\mathrm{LF}_{2,1}[m_\Phi]-\\
&\frac{1}{108}\,g_1^2\left(3\,s_\gamma^2\,\overline{y_e}^{pi3}\,y_e^{pi4}\,\delta_{i1i2}+\left(9\,s_\gamma^2\,\overline{y_d}^{i2p}\,y_d^{i1p}-9\,c_\gamma^2\,\overline{y_u}^{i2p}\,y_u^{i1p}+4\,g_1^2\,\delta_{i1i2}\right)\delta_{i3i4}\right)\\
&\quad\mathrm{LF}_{3,0}[m_\Phi]+\frac{5}{72}\,g_1^4\,\mathrm{LF}_{4,-1}[m_\Phi]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{4}{135}\,g_1^4\,\mathrm{LF}_{5,-2}[m_\Phi]\,\delta_{i1i2}\,\delta_{i3i4}-\\
&\frac{1}{27}\,g_1^4\,\mathrm{LF}_{3,0}[\tilde\mu]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{18}\,g_1^4\,\mathrm{LF}_{4,-1}[\tilde\mu]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{8}{135}\,g_1^4\,\mathrm{LF}_{5,-2}[\tilde\mu]\,\delta_{i1i2}\,\delta_{i3i4}-\\
&\frac{1}{36}\,g_1^4\,\mathrm{LF}_{2,1,0}\left[m_1,\,m_{\tilde e}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{36}\,g_1^4\,\mathrm{LF}_{2,2,-1}\left[m_1,\,m_{\tilde e}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4}+\\
&\frac{1}{18}\,g_1^4\,\mathrm{LF}_{3,1,-1}\left[m_1,\,m_{\tilde e}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{36}\,g_1^4\,\mathrm{LF}_{4,1,-2}\left[m_1,\,m_{\tilde e}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4}-\\
&\frac{1}{36}\,g_1^4\,\mathrm{LF}_{2,1,0}\left[m_1,\,m_{\tilde e}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{36}\,g_1^4\,\mathrm{LF}_{2,2,-1}\left[m_1,\,m_{\tilde e}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4}+\\
&\frac{1}{18}\,g_1^4\,\mathrm{LF}_{3,1,-1}\left[m_1,\,m_{\tilde e}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{36}\,g_1^4\,\mathrm{LF}_{4,1,-2}\left[m_1,\,m_{\tilde e}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4}-\\
&\frac{1}{1296}\,g_1^4\,\mathrm{LF}_{2,1,0}\left[m_1,\,m_{\tilde q}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{1296}\,g_1^4\,\mathrm{LF}_{2,2,-1}\left[m_1,\,m_{\tilde q}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4}+\\
&\frac{1}{648}\,g_1^4\,\mathrm{LF}_{3,1,-1}\left[m_1,\,m_{\tilde q}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{1296}\,g_1^4\,\mathrm{LF}_{4,1,-2}\left[m_1,\,m_{\tilde q}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4}-\\
&\frac{1}{1296}\,g_1^4\,\mathrm{LF}_{2,1,0}\left[m_1,\,m_{\tilde q}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{1296}\,g_1^4\,\mathrm{LF}_{2,2,-1}\left[m_1,\,m_{\tilde q}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4}+\\
&\frac{1}{648}\,g_1^4\,\mathrm{LF}_{3,1,-1}\left[m_1,\,m_{\tilde q}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{1296}\,g_1^4\,\mathrm{LF}_{4,1,-2}\left[m_1,\,m_{\tilde q}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4}-\\
&\frac{1}{48}\,g_1^2\,g_2^2\,\mathrm{LF}_{2,1,0}\left[m_2,\,m_{\tilde q}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{48}\,g_1^2\,g_2^2\,\mathrm{LF}_{2,2,-1}\left[m_2,\,m_{\tilde q}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4}+\\
&\frac{1}{24}\,g_1^2\,g_2^2\,\mathrm{LF}_{3,1,-1}\left[m_2,\,m_{\tilde q}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{48}\,g_1^2\,g_2^2\,\mathrm{LF}_{4,1,-2}\left[m_2,\,m_{\tilde q}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4}-\\
&\frac{1}{48}\,g_1^2\,g_2^2\,\mathrm{LF}_{2,1,0}\left[m_2,\,m_{\tilde q}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{48}\,g_1^2\,g_2^2\,\mathrm{LF}_{2,2,-1}\left[m_2,\,m_{\tilde q}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4}+\\
&\frac{1}{24}\,g_1^2\,g_2^2\,\mathrm{LF}_{3,1,-1}\left[m_2,\,m_{\tilde q}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{48}\,g_1^2\,g_2^2\,\mathrm{LF}_{4,1,-2}\left[m_2,\,m_{\tilde q}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4}-\\
&\frac{1}{27}\,g_1^2\,g_3^2\,\mathrm{LF}_{2,1,0}\left[m_3,\,m_{\tilde q}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{27}\,g_1^2\,g_3^2\,\mathrm{LF}_{2,2,-1}\left[m_3,\,m_{\tilde q}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4}+\\
&\frac{2}{27}\,g_1^2\,g_3^2\,\mathrm{LF}_{3,1,-1}\left[m_3,\,m_{\tilde q}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{27}\,g_1^2\,g_3^2\,\mathrm{LF}_{4,1,-2}\left[m_3,\,m_{\tilde q}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4}-\\
&\frac{1}{27}\,g_1^2\,g_3^2\,\mathrm{LF}_{2,1,0}\left[m_3,\,m_{\tilde q}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{27}\,g_1^2\,g_3^2\,\mathrm{LF}_{2,2,-1}\left[m_3,\,m_{\tilde q}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4}+\\
&\frac{2}{27}\,g_1^2\,g_3^2\,\mathrm{LF}_{3,1,-1}\left[m_3,\,m_{\tilde q}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{27}\,g_1^2\,g_3^2\,\mathrm{LF}_{4,1,-2}\left[m_3,\,m_{\tilde q}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4}-\\
&\frac{1}{9}\,g_1^2\,\overline{y_d}^{i2p}\,y_d^{i1p}\,\mathrm{LF}_{2,1,0}\left[m_{\tilde d}^p,\,\tilde\mu\right]\delta_{i3i4}+\frac{1}{18}\,g_1^2\,\overline{y_d}^{i2p}\,y_d^{i1p}\,\mathrm{LF}_{2,2,-1}\left[m_{\tilde d}^p,\,\tilde\mu\right]\delta_{i3i4}+\\
&\frac{1}{18}\,g_1^2\,\overline{y_d}^{i2p}\,y_d^{i1p}\,\mathrm{LF}_{3,1,-1}\left[m_{\tilde d}^p,\,\tilde\mu\right]\delta_{i3i4}+\frac{1}{18}\,g_1^4\,\mathrm{LF}_{2,1,0}\left[m_{\tilde e}^{i3},\,m_1\right]\delta_{i1i2}\,\delta_{i3i4}-\\
&\frac{1}{36}\,g_1^4\,\mathrm{LF}_{3,1,-1}\left[m_{\tilde e}^{i3},\,m_1\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{18}\,g_1^4\,\mathrm{LF}_{2,1,0}\left[m_{\tilde e}^{i4},\,m_1\right]\delta_{i1i2}\,\delta_{i3i4}-\\
&\frac{1}{36}\,g_1^4\,\mathrm{LF}_{3,1,-1}\left[m_{\tilde e}^{i4},\,m_1\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{18}\,g_1^2\,\overline{y_e}^{pi3}\,y_e^{pi4}\,\mathrm{LF}_{2,1,0}\left[m_{\tilde l}^p,\,\tilde\mu\right]\delta_{i1i2}-\\
&\frac{1}{36}\,g_1^2\,\overline{y_e}^{pi3}\,y_e^{pi4}\,\mathrm{LF}_{2,2,-1}\left[m_{\tilde l}^p,\,\tilde\mu\right]\delta_{i1i2}-\frac{1}{36}\,g_1^2\,\overline{y_e}^{pi3}\,y_e^{pi4}\,\mathrm{LF}_{3,1,-1}\left[m_{\tilde l}^p,\,\tilde\mu\right]\delta_{i1i2}+\\
&\frac{1}{648}\,g_1^4\,\mathrm{LF}_{2,1,0}\left[m_{\tilde q}^{i1},\,m_1\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{1296}\,g_1^4\,\mathrm{LF}_{3,1,-1}\left[m_{\tilde q}^{i1},\,m_1\right]\delta_{i1i2}\,\delta_{i3i4}+\\
&\frac{1}{24}\,g_1^2\,g_2^2\,\mathrm{LF}_{2,1,0}\left[m_{\tilde q}^{i1},\,m_2\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{48}\,g_1^2\,g_2^2\,\mathrm{LF}_{3,1,-1}\left[m_{\tilde q}^{i1},\,m_2\right]\delta_{i1i2}\,\delta_{i3i4}+\\
&\frac{2}{27}\,g_1^2\,g_3^2\,\mathrm{LF}_{2,1,0}\left[m_{\tilde q}^{i1},\,m_3\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{27}\,g_1^2\,g_3^2\,\mathrm{LF}_{3,1,-1}\left[m_{\tilde q}^{i1},\,m_3\right]\delta_{i1i2}\,\delta_{i3i4}+\\
&\frac{1}{648}\,g_1^4\,\mathrm{LF}_{2,1,0}\left[m_{\tilde q}^{i2},\,m_1\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{1296}\,g_1^4\,\mathrm{LF}_{3,1,-1}\left[m_{\tilde q}^{i2},\,m_1\right]\delta_{i1i2}\,\delta_{i3i4}+\\
&\frac{1}{24}\,g_1^2\,g_2^2\,\mathrm{LF}_{2,1,0}\left[m_{\tilde q}^{i2},\,m_2\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{48}\,g_1^2\,g_2^2\,\mathrm{LF}_{3,1,-1}\left[m_{\tilde q}^{i2},\,m_2\right]\delta_{i1i2}\,\delta_{i3i4}+\\
&\frac{2}{27}\,g_1^2\,g_3^2\,\mathrm{LF}_{2,1,0}\left[m_{\tilde q}^{i2},\,m_3\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{27}\,g_1^2\,g_3^2\,\mathrm{LF}_{3,1,-1}\left[m_{\tilde q}^{i2},\,m_3\right]\delta_{i1i2}\,\delta_{i3i4}+\\
&\frac{2}{9}\,g_1^2\,\overline{y_u}^{i2p}\,y_u^{i1p}\,\mathrm{LF}_{2,1,0}\left[m_{\tilde u}^p,\,\tilde\mu\right]\delta_{i3i4}-\frac{1}{9}\,g_1^2\,\overline{y_u}^{i2p}\,y_u^{i1p}\,\mathrm{LF}_{2,2,-1}\left[m_{\tilde u}^p,\,\tilde\mu\right]\delta_{i3i4}-\\
&\frac{1}{9}\,g_1^2\,\overline{y_u}^{i2p}\,y_u^{i1p}\,\mathrm{LF}_{3,1,-1}\left[m_{\tilde u}^p,\,\tilde\mu\right]\delta_{i3i4}+\frac{1}{18}\,g_1^2\,\overline{y_d}^{i2p}\,y_d^{i1p}\,\mathrm{LF}_{2,1,0}\left[\tilde\mu,\,m_{\tilde d}^p\right]\delta_{i3i4}+\\
&\frac{5}{36}\,g_1^2\,\overline{y_d}^{i2p}\,y_d^{i1p}\,\mathrm{LF}_{3,1,-1}\left[\tilde\mu,\,m_{\tilde d}^p\right]\delta_{i3i4}-\frac{1}{36}\,g_1^2\,\overline{y_d}^{i2p}\,y_d^{i1p}\,\mathrm{LF}_{4,1,-2}\left[\tilde\mu,\,m_{\tilde d}^p\right]\delta_{i3i4}-\\
&\frac{1}{36}\,g_1^2\,\overline{y_e}^{pi3}\,y_e^{pi4}\,\mathrm{LF}_{2,1,0}\left[\tilde\mu,\,m_{\tilde l}^p\right]\delta_{i1i2}+\frac{5}{36}\,g_1^2\,\overline{y_e}^{pi3}\,y_e^{pi4}\,\mathrm{LF}_{3,1,-1}\left[\tilde\mu,\,m_{\tilde l}^p\right]\delta_{i1i2}-\\
&\frac{1}{18}\,g_1^2\,\overline{y_e}^{pi3}\,y_e^{pi4}\,\mathrm{LF}_{4,1,-2}\left[\tilde\mu,\,m_{\tilde l}^p\right]\delta_{i1i2}-\frac{1}{9}\,g_1^2\,\overline{y_u}^{i2p}\,y_u^{i1p}\,\mathrm{LF}_{2,1,0}\left[\tilde\mu,\,m_{\tilde u}^p\right]\delta_{i3i4}-\\
&\frac{1}{36}\,g_1^2\,\overline{y_u}^{i2p}\,y_u^{i1p}\,\mathrm{LF}_{3,1,-1}\left[\tilde\mu,\,m_{\tilde u}^p\right]\delta_{i3i4}-\frac{1}{36}\,g_1^2\,\overline{y_u}^{i2p}\,y_u^{i1p}\,\mathrm{LF}_{4,1,-2}\left[\tilde\mu,\,m_{\tilde u}^p\right]\delta_{i3i4}-\\
&\frac{1}{72}\,g_1^4\,\mathrm{LF}_{2,1,1,-1}\left[m_1,\,m_{\tilde e}^{i3},\,m_{\tilde q}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{36}\,g_1^4\,m_1^2\,\mathrm{LF}_{2,1,1,0}\left[m_1,\,m_{\tilde e}^{i3},\,m_{\tilde q}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4}-\\
&\frac{1}{72}\,g_1^4\,\mathrm{LF}_{2,1,1,-1}\left[m_1,\,m_{\tilde e}^{i4},\,m_{\tilde q}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{36}\,g_1^4\,m_1^2\,\mathrm{LF}_{2,1,1,0}\left[m_1,\,m_{\tilde e}^{i4},\,m_{\tilde q}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4}-\\
&\frac{1}{4}\,\overline{y_d}^{i2p}\,y_d^{i1p}\,\overline{y_e}^{ri3}\,y_e^{ri4}\,\mathrm{LF}_{2,1,1,-1}\left[\tilde\mu,\,m_{\tilde d}^p,\,m_{\tilde l}^r\right]-\\
&\left.\frac{1}{2}\,\tilde\mu^2\,\overline{y_e}^{pi3}\,y_e^{pi4}\,\overline{y_u}^{i2r}\,y_u^{i1r}\,\mathrm{LF}_{2,1,1,0}\left[\tilde\mu,\,m_{\tilde l}^p,\,m_{\tilde u}^r\right]\right)
\end{aligned}
$$